CS-216-Chain_Snatchers-Bitcoin-Transaction-simulation Report

Introduction

This project demonstrates the creation, signing, broadcasting, and verification of Bitcoin transactions using Bitcoin Core in regtest mode. The objective of this simulation is to understand how Bitcoin transactions work internally by manually constructing and analyzing both Legacy (P2PKH) and SegWit (P2SH-P2WPKH) transactions. Using RPC commands, Python scripts, and the btcdeb Bitcoin Script debugger, we simulate the complete transaction workflow including wallet funding, raw transaction creation, script validation, and UTXO spending. The project also compares Legacy and SegWit transactions in terms of script structure, transaction size, weight, and fees, providing a clear understanding of how SegWit improves efficiency and solves issues such as transaction malleability. This simulation helps in gaining practical knowledge of Bitcoin's scripting system, transaction format, and validation mechanism at a low level.

Team Information

Team Name: Chain Snatchers

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Requirements:

- Bitcoin Core (bitcoind version > 27.0 & bitcoin-cli)
- Python (version>3.8)
- Libraries: Bitcoinrpc
- Bitcoin Debugger: btcdeb

Initialization

Configuration

We first set up Bitcoin Core using the setup wizard and edited the configuration file to meet the project's requirements.

```
regtest=1
server=1
txindex=1

[regtest]
rpcuser=bitcoinrpc
```

```
rpcpassword=chainsnatcher1
rpcallowip=127.0.0.1
rpcport=18443

debug=validation

paytxfee=0.0001
fallbackfee=0.0002
mintxfee=0.00001
txconfirmtarget=6
```

Once this is done, we can start with the making of the required transaction

Setup

- Run the bitcoind in regtest mode using `bitcoind -regtest`
- Establish RPC connection with Python file
- Create a wallet and connect via the RPC
- Fund the Wallet by mining 101 Blocks (Standard), generating 50 BTC

Download all the Python files locally and keep the bitcoin.conf file in the Bitcoin running directory. Run the Legacy_work.py for legacy transactions and segwit_work.py for segwit transactions

Note: The wallet is unloaded after every execution of the file to avoid conflict between the wallets of both files. But if the same file were to be run more than once, the entire regtest is required to be manually reset by deleting the wallets and regtest folders.

Part 1: Legacy Transaction

Workflow:

- The wallet first sends 40 BTC to address A to fund it.
- A raw transaction is then created to send half of A's balance to address B.
- This transaction is signed and broadcast to the network, producing the transaction ID (txid) for the A → B transaction.
- The output of this transaction becomes a UTXO belonging to address B.
- The program retrieves this UTXO using listunspent.
- The retrieved UTXO contains the txid and output index (vout) of the A → B transaction.
- This UTXO is then used as the input for the next raw transaction, where B sends funds to address C.
- When the B → C transaction is decoded, the previous transaction ID is obtained using: `decoded_spending_tx["vin"][0]["txid"]`
- This value corresponds to the A → B transaction ID, demonstrating that the B → C transaction spends the output created in the previous transaction.

Script Analysis

- Using the btcdeb, we verify the transaction's validity
- We are using `./btcdeb <ScriptSig> <ScriptPubKey>`

1. Locking Script Creation (ScriptPubKey)

```
When A sends BTC to B, the output contains a ScriptPubKey.

Structure:

  OP_DUP OP_HASH160 <PubKeyHash> OP_EQUALVERIFY OP_CHECKSIG
  <PubKeyHash>-> is the hash of B's public key.

This serves as the challenge script, defining the conditions for spendi
ng the output.
```

2. Unlocking Script Creation (ScriptSig)

```
When B spends the output (B → C), the input contains a ScriptSig.

Structure:

  <signature> <public_key>
The signature proves ownership of the private key corresponding to the
public key.
```

3. Transaction Linking

```
The B → C transaction input references the previous transaction using:
txid-> ID of the A → B transaction
vout-> index of the output being spent
```

Script Execution using btcdeb

The entire execution takes place using a stack-based execution process :

- Initial stack -> []
- Push Signature -> [Sig]
- Push PubKey -> [Sig, PubKey]
- OP_DUP -> [Sig, PubKey, PubKey]
- OP_HASH160 -> [Sig, PubKey, HASH160(PubKey)]
- Push PubKeyHash -> [Sig, PubKey, HASH160(PubKey), PubKeyHash]
- OP_EQUALVERIFY -> [Sig, PubKey]
- OP_CHECKSIG -> [TRUE]

Images:

Outputs:

```
----------------------------- Decoded Scripts -------------------------------------
Transaction from A -> B :  {'txid': '6e81079977396dd19594cbc0a91b906a96d7e3f9e4265af14473698395fb98e6', 'hash': '6e81079977396dd19594cbc0a91b906a96d7e3f9e4265af144
73698395fb98e6', 'version': 2, 'size': 225, 'vsize': 225, 'weight': 900, 'locktime': 0, 'vin': [{'txid': '5cb2c27a376a9769d63e6dd53811ad3d4cfbf1bb84877ae2b9b1f1d1a
7b40f6f', 'vout': 1, 'scriptSig': {'asm': '304402202224e247bacc8de00c0febc1f64e51614f5c6c84459b241ee53fcd8e16d08f53022041a18eea3107c8f209159fc2368728de96454ab37138
993c27db7dcd0793a0ff[ALL] 026e8267b4614ef7be10f70df6f827637e041fc7a978d3c6f5f7bad05ff52722db', 'hex': '47304402202224e247bacc8de00c0febc1f64e51614f5c6c84459b241ee5
3fcd8e16d08f53022041a18eea3107c8f209159fc2368728de96454ab37138993c27db7dcd0793a0ff0121026e8267b4614ef7be10f70df6f827637e041fc7a978d3c6f5f7bad05ff52722db'}, 'sequen
ce': 4294967293}], 'vout': [{'value': Decimal('20.00000000'), 'n': 0, 'scriptPubKey': {'asm': 'OP_DUP OP_HASH160 b779be9d5b87479295c3e303dbed541636552e9e OP_EQUALV
ERIFY OP_CHECKSIG', 'desc': 'addr(mxF5o7zJR5BoFVbShyo9C91TLcvB6Yx1nm)#6f0zrv85', 'hex': '76a914b779be9d5b87479295c3e303dbed541636552e9e88ac', 'address': 'mxF5o7zJR
5BoFVbShyo9C91TLcvB6Yx1nm', 'type': 'pubkeyhash'}}, {'value': Decimal('19.99990000'), 'n': 1, 'scriptPubKey': {'asm': 'OP_DUP OP_HASH160 f533ac72b6dac810f65c9572e8
5c258637e4d5b4 OP_EQUALVERIFY OP_CHECKSIG', 'desc': 'addr(n3sTkQ6BjGZcpMGQm2WhKe9CNnTMG1FBCg)#2umza06x', 'hex': '76a914f533ac72b6dac810f65c9572e85c258637e4d5b488ac
', 'address': 'n3sTkQ6BjGZcpMGQm2WhKe9CNnTMG1FBCg', 'type': 'pubkeyhash'}}]}
Transaction from B -> C :  {'txid': '6f6b65d5d655f737c987564faeb568ed9593d3a945f5f4e75eefba24fe3e1acc', 'hash': '6f6b65d5d655f737c987564faeb568ed9593d3a945f5f4e75e
efba24fe3e1acc', 'version': 2, 'size': 225, 'vsize': 225, 'weight': 900, 'locktime': 0, 'vin': [{'txid': '6e81079977396dd19594cbc0a91b906a96d7e3f9e4265af1447369839
5fb98e6', 'vout': 0, 'scriptSig': {'asm': '304402202b7c490d918d6645c31e0f9af51b6dcb51975ee7de1a6ff1dd11749bd9b4c6d702206642fcb392bb3fd15f15c34ad29e989f356bce03035a
74bcb8c54dd474bd4da6[ALL] 031373e67a49ab951c75eac9fba0940c6bae541319eecdd39f64462b7ec64951f6', 'hex': '47304402202b7c490d918d6645c31e0f9af51b6dcb51975ee7de1a6ff1dd
11749bd9b4c6d702206642fcb392bb3fd15f15c34ad29e989f356bce03035a74bcb8c54dd474bd4da60121031373e67a49ab951c75eac9fba0940c6bae541319eecdd39f64462b7ec64951f6'}, 'sequen
ce': 4294967293}], 'vout': [{'value': Decimal('10.00000000'), 'n': 0, 'scriptPubKey': {'asm': 'OP_DUP OP_HASH160 ad011b0c12441f8ea7182eb2258106bddcfb16b2 OP_EQUALV
ERIFY OP_CHECKSIG', 'desc': 'addr(mwHiX4bW6x45k8HkptLfPJbYThSarUpjHN)#gfqws8p9', 'hex': '76a914ad011b0c12441f8ea7182eb2258106bddcfb16b288ac', 'address': 'mwHiX4bW6
x45k8HkptLfPJbYThSarUpjHN', 'type': 'pubkeyhash'}}, {'value': Decimal('9.99990000'), 'n': 1, 'scriptPubKey': {'asm': 'OP_DUP OP_HASH160 b779be9d5b87479295c3e303dbe
d541636552e9e OP_EQUALVERIFY OP_CHECKSIG', 'desc': 'addr(mxF5o7zJR5BoFVbShyo9C91TLcvB6Yx1nm)#6f0zrv85', 'hex': '76a914b779be9d5b87479295c3e303dbed541636552e9e88ac'
, 'address': 'mxF5o7zJR5BoFVbShyo9C91TLcvB6Yx1nm', 'type': 'pubkeyhash'}}]}
----------------------------- Transaction IDs -------------------------------------
Transaction TD from A -> B :  6e81079977396dd19594cbc0a91b906a96d7e3f9e4265af14473698395fb98e6
Transaction ID from B -> C :  6f6b65d5d655f737c987564faeb568ed9593d3a945f5f4e75eefba24fe3e1acc
----------------------------- Required Details ------------------------------------
Transaction ID for B -> C: 6f6b65d5d655f737c987564faeb568ed9593d3a945f5f4e75eefba24fe3e1acc
ScriptSig (response provided by B): 304402202b7c490d918d6645c31e0f9af51b6dcb51975ee7de1a6ff1dd11749bd9b4c6d702206642fcb392bb3fd15f15c34ad29e989f356bce03035a74bcb8c
54dd474bd4da6[ALL] 031373e67a49ab951c75eac9fba0940c6bae541319eecdd39f64462b7ec64951f6
Challenge Script (ScriptPubKey from A->B tx): OP_DUP OP_HASH160 b779be9d5b87479295c3e303dbed541636552e9e OP_EQUALVERIFY OP_CHECKSIG
```

Legacy 1

```
----------------------------Transaction Sizes--------------------------------
 A->B Transaction
Transaction Size : 225
Transaction Virtual size: 225
Transaction Weight: 900
Transaction Size (in bytes): 225.0
 B->C Transaction
Transaction Size : 225
Transaction Virtual size: 225
Transaction Weight: 900
Transaction Size (in bytes): 225.0
----------------------------- Final Balances --------------------------------------
DEBUG:BitcoinRPC:-34-> listunspent [1, 9999999, ["n3sTkQ6BjGZcpMGQm2WhKe9CNnTMG1FBCg"]]
DEBUG:BitcoinRPC:<-34- [{"txid": "6e81079977396dd19594cbc0a91b906a96d7e3f9e4265af14473698395fb98e6", "vout": 1, "address": "n3sTkQ6BjGZcpMGQm2WhKe9CNnTMG1FBCg", "l
abel": "A", "scriptPubKey": "76a914f533ac72b6dac810f65c9572e85c258637e4d5b488ac", "amount": 19.9999, "confirmations": 2, "spendable": true, "solvable": true, "desc
": "pkh([91de3fd5/44h/1h/0h/0/1]026e8267b4614ef7be10f70df6f827637e041fc7a978d3c6f5f7bad05ff52722db)#fu60fjvy", "parent_descs": ["pkh([91de3fd5/44h/1h/0h]tpubDDsvz4
1Hn2xuLunq9ZVFAcyCjXqyyvUHsbusaE5z3PShRrkX2Sgw2ouW8nXurErYJMbRLxwwPnGUKkrV3F9LSn5w7TyRPCQSwNdMNNUTXG9/0/*)#vgtrf450"], "safe": true}]
Balance of n3sTkQ6BjGZcpMGQm2WhKe9CNnTMG1FBCg: 19.99990000
DEBUG:BitcoinRPC:-35-> listunspent [1, 9999999, ["mxF5o7zJR5BoFVbShyo9C91TLcvB6Yx1nm"]]
DEBUG:BitcoinRPC:<-35- [{"txid": "6f6b65d5d655f737c987564faeb568ed9593d3a945f5f4e75eefba24fe3e1acc", "vout": 1, "address": "mxF5o7zJR5BoFVbShyo9C91TLcvB6Yx1nm", "l
abel": "B", "scriptPubKey": "76a914b779be9d5b87479295c3e303dbed541636552e9e88ac", "amount": 9.9999, "confirmations": 1, "spendable": true, "solvable": true, "desc"
: "pkh([91de3fd5/44h/1h/0h/0/2]031373e67a49ab951c75eac9fba0940c6bae541319eecdd39f64462b7ec64951f6)#6pzf09tg", "parent_descs": ["pkh([91de3fd5/44h/1h/0h]tpubDDsvz41
Hn2xuLunq9ZVFAcyCjXqyyvUHsbusaE5z3PShRrkX2Sgw2ouW8nXurErYJMbRLxwwPnGUKkrV3F9LSn5w7TyRPCQSwNdMNNUTXG9/0/*)#vgtrf450"], "safe": true}]
Balance of mxF5o7zJR5BoFVbShyo9C91TLcvB6Yx1nm: 9.99990000
DEBUG:BitcoinRPC:-36-> listunspent [1, 9999999, ["mwHiX4bW6x45k8HkptLfPJbYThSarUpjHN"]]
DEBUG:BitcoinRPC:<-36- [{"txid": "6f6b65d5d655f737c987564faeb568ed9593d3a945f5f4e75eefba24fe3e1acc", "vout": 0, "address": "mwHiX4bW6x45k8HkptLfPJbYThSarUpjHN", "l
abel": "C", "scriptPubKey": "76a914ad011b0c12441f8ea7182eb2258106bddcfb16b288ac", "amount": 10.0, "confirmations": 1, "spendable": true, "solvable": true, "desc":
"pkh([91de3fd5/44h/1h/0h/0/3]023f57a27773f7c2c2afa717621e9056f30e006dcd5c81e560bdeb909f9f9be8f5)#ar647ly2", "parent_descs": ["pkh([91de3fd5/44h/1h/0h]tpubDDsvz41Hn
2xuLunq9ZVFAcyCjXqyyvUHsbusaE5z3PShRrkX2Sgw2ouW8nXurErYJMbRLxwwPnGUKkrV3F9LSn5w7TyRPCQSwNdMNNUTXG9/0/*)#vgtrf450"], "safe": true}]
Balance of mwHiX4bW6x45k8HkptLfPJbYThSarUpjHN: 10.00000000
Balance of A: 19.99990000
Balance of B: 9.99990000
Balance of C: 10.00000000
```

legacy 2

Verification:

```
sandeep@sandeep-IdeaPad-Slim-3-15IRH8: /bitcoin/src/bitcoin$ btcdeb --tx=0200000015dada0f490a03e2d1b5c55e54fb3e6a23e99be0d248cc1a03acf4bfd6c5d91bf000000006a47304022026c13354a1a700e18e61753b96003cf77fc34c
1b428fd4046fb3face604514c102202346fdae1d5da68d045531d7406b72952dd7a32eae2a47a9295ec12947b67e9d012102882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20ffdffffff0200ca9a3b000000001976a91454dc22
ca0f0921897a133370596f0c93a6577eb88acf0a29a3b000000001976a91481a7c0560a550560bc542c3daea5287174770289 88ac00000000 --txin=0200000012f9cccc0dd5fb230ecfdfcb1ef44306c43973a889b914e0a8b864022eafeb3f001000000
6a47304402207ddaf0abd2dccff4e587b948088f4b51889e9556afe788114fb93e693f48231f0220553d3520a76e6ac94c03110e108ab8d25872bd9c716de42259f8ea9b744e55660121030c8b74b96e449 8bb1ee2bb4c9194e97fca29dddb4120e1cf374a6f
55a8611e1bfdffffff02009435770000000001976a91481a7c0560a550560bc542c3daea5287174770289 88acf06c357700000000001976a914070fb9cd3842942459 5a082d8ed57ab43df7c43e88ac00000000
btcdeb 5.0.24 -- type `btcdeb -h` for start up options
LOG: signing segwit taproot
notice: btcdeb has gotten quieter; use --verbose if necessary (this message is temporary)
input tx index = 0; tx input vout = 0; value = 2000000000
got witness stack of size 0
8 op script loaded. type `help` for usage information
script                                                      |  stack
------------------------------------------------------------+--------
3044022026c13354a1a700e18e61753b96003cf77fc34c1b428fd4046fb3fac... |
02882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20f |
<<< scriptPubKey >>>                                        |
OP_DUP                                                      |
OP_HASH160                                                  |
81a7c0560a550560bc542c3daea5287174770289                    |
OP_EQUALVERIFY                                              |
OP_CHECKSIG                                                 |
#0000 3044022026c13354a1a700e18e61753b96003cf77fc34c1b428fd4046fb3face604514c102202346fdae1d5da68d045531d7406b72952dd7a32eae2a47a9295ec12947b67e9d01
btcdeb> step
                <> PUSH stack 3044022026c13354a1a700e18e61753b96003cf77fc34c1b428fd4046fb3face604514c102202346fdae1d5da68d045531d7406b72952dd7a32eae2a47a9295ec12947b67e9d01
script                                                      |                                                       stack
------------------------------------------------------------+-------------------------------------------------------
02882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20f | 3044022026c13354a1a700e18e61753b96003cf77fc34c1b428fd4046fb3fac...
<<< scriptPubKey >>>                                        |
OP_DUP                                                      |
OP_HASH160                                                  |
81a7c0560a550560bc542c3daea5287174770289                    |
OP_EQUALVERIFY                                              |
OP_CHECKSIG                                                 |
#0001 02882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20f
btcdeb> step
                <> PUSH stack 02882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20f
script                                                      |                                                       stack
------------------------------------------------------------+-------------------------------------------------------
<<< scriptPubKey >>>                                        | 02882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20f
OP_DUP                                                      | 3044022026c13354a1a700e18e61753b96003cf77fc34c1b428fd4046fb3fac...
OP_HASH160                                                  |
81a7c0560a550560bc542c3daea5287174770289                    |
OP_EQUALVERIFY                                              |
OP_CHECKSIG                                                 |
<<< scriptPubKey >>>
btcdeb> step
script                                                      |                                                       stack
------------------------------------------------------------+-------------------------------------------------------
OP_DUP                                                      | 02882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20f
OP_HASH160                                                  | 3044022026c13354a1a700e18e61753b96003cf77fc34c1b428fd4046fb3fac...
81a7c0560a550560bc542c3daea5287174770289                    |
OP_EQUALVERIFY                                              |
OP_CHECKSIG                                                 |
```

Legacy 1

```
btcdeb> step
                <> PUSH stack 02882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20f
script                                                      |                                                       stack
------------------------------------------------------------+-------------------------------------------------------
OP_HASH160                                                  | 02882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20f
81a7c0560a550560bc542c3daea5287174770289                    | 02882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20f
OP_EQUALVERIFY                                              | 3044022026c13354a1a700e18e61753b96003cf77fc34c1b428fd4046fb3fac...
OP_CHECKSIG                                                 |
#0004 OP_HASH160
btcdeb> step
                <> POP  stack
                <> PUSH stack 81a7c0560a550560bc542c3daea5287174770289
script                                                      |                                                       stack
------------------------------------------------------------+-------------------------------------------------------
81a7c0560a550560bc542c3daea5287174770289                    |                                 81a7c0560a550560bc542c3daea5287174770289
OP_EQUALVERIFY                                              | 02882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20f
OP_CHECKSIG                                                 | 3044022026c13354a1a700e18e61753b96003cf77fc34c1b428fd4046fb3fac...
#0005 81a7c0560a550560bc542c3daea5287174770289
btcdeb> step
                <> PUSH stack 81a7c0560a550560bc542c3daea5287174770289
script                                                      |                                                       stack
------------------------------------------------------------+-------------------------------------------------------
OP_EQUALVERIFY                                              |                                 81a7c0560a550560bc542c3daea5287174770289
OP_CHECKSIG                                                 |                                 81a7c0560a550560bc542c3daea5287174770289
                                                            | 02882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20f
                                                            | 3044022026c13354a1a700e18e61753b96003cf77fc34c1b428fd4046fb3fac...
#0006 OP_EQUALVERIFY
```

Legacy 2

```
btcdeb> step
                <> POP  stack
                <> POP  stack
                <> PUSH stack 01
                <> POP  stack
script                                                      |                                                       stack
------------------------------------------------------------+-------------------------------------------------------
OP_CHECKSIG                                                 | 02882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20f
                                                            | 3044022026c13354a1a700e18e61753b96003cf77fc34c1b428fd4046fb3fac...

#0007 OP_CHECKSIG
btcdeb> step
EvalChecksig() sigversion=0
Eval Checksig Pre-Tapscript
GenericTransactionSignatureChecker::CheckECDSASignature(71 len sig, 33 len pubkey, sigversion=0)
  sig     = 3044022026c13354a1a700e18e61753b96003cf77fc34c1b428fd4046fb3face604514c102202346fdae1d5da68d045531d7406b72952dd7a32eae2a47a9295ec12947b67e9d01
  pub key     = 02882f1acc8d104e398e2075cc921f9e9e719d54d2d1c5975075f3ae3c2583f20f
  script code = 76a91481a7c0560a550560bc542c3daea5287174770289 88ac
  hash type   = 01 (SIGHASH_ALL)
SignatureHash(nIn=0, nHashType=01, amount=2000000000)
- sigversion = SIGVERSION_BASE (non-segwit style)
 << TxTo.vin[nInput=0].prevout = COutPoint(bf915d6cfd, 0)
(SerializeScriptCode)
 << scriptCode.size()=25 - nCodeSeparators=0
 << script:76a91481a7c0560a550560bc542c3daea5287174770289 88ac
 << TxTo.vin[nInput].nSequence = 4294967293 [0xfffffffd]
  sighash     = 85415db5e1d9a8690fae79aa2b9d8f4c05c3c9275c2c2260e867aebfd6f6bf7c
  pubkey.VerifyECDSASignature(sig=3044022026c13354a1a700e18e61753b96003cf77fc34c1b428fd4046fb3face604514c102202346fdae1d5da68d045531d7406b72952dd7a32eae2a47a9295ec12947b67e9d, sighash=85415db5e1d9a8690fae
79aa2b9d8f4c05c3c9275c2c2260e867aebfd6f6bf7c):
  result: success
                <> POP  stack
                <> POP  stack
                <> PUSH stack 01
script                                                      |                                                       stack
------------------------------------------------------------+-------------------------------------------------------
                                                            |                                                            01

btcdeb> step
script                                                      |                                                       stack
------------------------------------------------------------+-------------------------------------------------------
                                                            |                                                            01
```

Legacy 3

Part 2: SegWit Transaction

Workflow:

- The wallet first sends 40 BTC to address A to fund it.
- A raw transaction is then created to send half of A's balance to address B.
- This transaction is signed and broadcast to the network, producing the transaction ID (txid) for the A → B transaction.
- The output of this transaction becomes a UTXO belonging to address B.
- The program retrieves this UTXO using listunspent.
- The retrieved UTXO contains the txid and output index (vout) of the A → B transaction.
- This UTXO is then used as the input for the next raw transaction, where B sends funds to address C.
- When the B → C transaction is decoded, the previous transaction ID is obtained using: `decoded_spending_tx["vin"][0]["txid"]`
- This value corresponds to the A → B transaction ID, demonstrating that the B → C transaction spends the output created in the previous transaction.

Script Analysis

- Using the btcdeb, we verify the transaction's validity
- We are using `./btcdeb <Witness> <RedeemScript / ScriptPubKey>`

1. Locking Script Creation (ScriptPubKey)

```
When A sends BTC to B using a SegWit address, the output contains a P2S
H ScriptPubKey.

Structure:

   OP_HASH160 <RedeemScriptHash> OP_EQUAL
   <RedeemScriptHash>-> is the HASH160 of the redeem script.

The redeem script corresponds to the SegWit witness program.
```

2. Redeem Script(Witness Program)

```
The redeem script embedded inside P2SH is:

   0 <PubKeyHash>

   0-> SegWit version number.

  <PubKeyHash>-> HASH160 of B's public key.

This script tells Bitcoin that the actual unlocking data will be stored
in the witness field.
```

3. Unlocking Script (ScriptSig)

```
For P2SH-P2WPKH, the ScriptSig contains only the redeem script:

    <redeem_script>

Example:

   0 <PubKeyHash>

So ScriptSig is much smaller than legacy ScriptSig.
```

4. Witness Data(txinwitness)

```
The actual unlocking data is placed in the witness field rather than th
e ScriptSig.

Witness structure:

  <signature>
  <public_key>

This is similar to the legacy ScriptSig, but moved into the witness.
```

5. Transaction Linking

```
The B->C transaction input references the previous output using:

   txid → transaction ID of A → B

   vout → output index

This output becomes the input for the next transaction.
```

Script Execution using btcdeb

The Entire Execution takes place using a stack-based execution process :

- Initial stack -> []
- Push Signature -> [Sig]
- Push PubKey -> [Sig, PubKey]
- Push 0 (from RedeemScript) -> [Sig, PubKey, 0]
- Push PubKeyHash -> [Sig, PubKey, 0, PubKeyHash]
- OP_HASH160 -> [Sig, PubKey, HASH160(RedeemScript)]
- Push ScriptHash -> [Sig, PubKey, HASH160(RedeemScript), ScriptHash]
- OP_EQUAL -> [Sig, PubKey]
- OP_DUP -> [Sig, PubKey, PubKey]
- OP_HASH160 -> [Sig, PubKey, HASH160(PubKey)]

- Push PubKeyHash -> [Sig, PubKey, HASH160(PubKey), PubKeyHash]
- OP_EQUALVERIFY -> [Sig, PubKey]
- OP_CHECKSIG -> [TRUE]

Images:

Outputs:

```
---------------------------- Decoded Scripts -----------------------------------
Decoded Scripts:
Transaction from A -> B :  {'txid': 'e9386e2fa42775a18fe3ae18c290a67f393c96e4b59e99da31519e5186bfd108', 'hash': '7dac89bc280235a9e66f081834a82174e4ceba51b340ac7307
825f62d3939824', 'version': 2, 'size': 247, 'vsize': 166, 'weight': 661, 'locktime': 0, 'vin': [{'txid': 'b94634cfb1d9caca62379b2fdd2efb867e439acb371430131cc2026b0
8145a5f', 'vout': 0, 'scriptSig': {'asm': '00145c7e9edd98581f4ac3127c43195bf53c6c09e9c6', 'hex': '1600145c7e9edd98581f4ac3127c43195bf53c6c09e9c6'}, 'txinwitness':
['304402200bd69522a47c32770d5d29d5fdf607117978ebd53d4c5a14d8925b612be6a8c30220657398c5f1121db8a9393915bef2fc8bf75da7c8d59b5b9c6384ee71685ee0d001', '020f67d1394038b
47013598f75b9c99c048c788e99773e8a5aa3b3c7675487465a'], 'sequence': 4294967293}], 'vout': [{'value': Decimal('20.00000000'), 'n': 0, 'scriptPubKey': {'asm': 'OP_HAS
H160 75fe0c248a1bb0f3e0280def4d79d55d351f74cb OP_EQUAL', 'desc': 'addr(2N417PFdwWNvRfwEXyCMM5fXYwSCv2yEShU)#upjwsjhd', 'hex': 'a91475fe0c248a1bb0f3e0280def4d79d55d
351f74cb87', 'address': '2N417PFdwWNvRfwEXyCMM5fXYwSCv2yEShU', 'type': 'scripthash'}}, {'value': Decimal('19.99990000'), 'n': 1, 'scriptPubKey': {'asm': 'OP_HASH16
0 11b8cc5383f7a8f977f2aeb640da862c6b80034d OP_EQUAL', 'desc': 'addr(2MtrvrYbc6F4NRDsarQV9NyVJUQoU9YyV7C)#hpy5z7jq', 'hex': 'a91411b8cc5383f7a8f977f2aeb640da862c6b8
0034d87', 'address': '2MtrvrYbc6F4NRDsarQV9NyVJUQoU9YyV7C', 'type': 'scripthash'}}]}
Transaction from B -> C :  {'txid': '9fce88f76f99e686bc3731fd4c1dbd761c19a8929ff37efcfa7ca00a8bd22d5a', 'hash': '288e9ac8bb4faf828a24b4d2c5d7548799896969c796952a1e
89688e902aeffd', 'version': 2, 'size': 247, 'vsize': 166, 'weight': 661, 'locktime': 0, 'vin': [{'txid': 'e9386e2fa42775a18fe3ae18c290a67f393c96e4b59e99da31519e518
6bfd108', 'vout': 0, 'scriptSig': {'asm': '001432fdaa8bb21a406be12967b880408cd09498de8d', 'hex': '16001432fdaa8bb21a406be12967b880408cd09498de8d'}, 'txinwitness':
['304402202cfd9ed3f7f95e028b92b49b4c81f81a48eb632d4ec395afae6d8f65afb2891c02200fdc9514c433ca4df875b09f4d7c6d48578bee1fc5636ba3834691b223134a6d01', '027babf4dffa0c3
b91b7ff39751bb3bfe35e6c1d80beb7de977b6152087880063f'], 'sequence': 4294967293}], 'vout': [{'value': Decimal('10.00000000'), 'n': 0, 'scriptPubKey': {'asm': 'OP_HAS
H160 6e13fe2f9930597324e9789b62ac27c224e5322d OP_EQUAL', 'desc': 'addr(2N3HGGWxTV3WKVcoxabiZRVKu83ojCAe6AC)#0sguqx50', 'hex': 'a9146e13fe2f9930597324e9789b62ac27c2
24e5322d87', 'address': '2N3HGGWxTV3WKVcoxabiZRVKu83ojCAe6AC', 'type': 'scripthash'}}, {'value': Decimal('9.99990000'), 'n': 1, 'scriptPubKey': {'asm': 'OP_HASH160
 75fe0c248a1bb0f3e0280def4d79d55d351f74cb OP_EQUAL', 'desc': 'addr(2N417PFdwWNvRfwEXyCMM5fXYwSCv2yEShU)#upjwsjhd', 'hex': 'a91475fe0c248a1bb0f3e0280def4d79d55d351f
74cb87', 'address': '2N417PFdwWNvRfwEXyCMM5fXYwSCv2yEShU', 'type': 'scripthash'}}]}
----------------------------- Transaction IDs ----------------------------------
Transaction TD from A -> B :  e9386e2fa42775a18fe3ae18c290a67f393c96e4b59e99da31519e5186bfd108
Transaction ID from B -> C :  9fce88f76f99e686bc3731fd4c1dbd761c19a8929ff37efcfa7ca00a8bd22d5a
```

Segwit 1

```
---------------------------- Required Details ----------------------------------
B -> C Transaction Hex: 0200000000010108d1bf86519e5131da999eb5e4963c397fa690c218aee38fa17527a42f6e38e900000000171600145c7e9edd98581f4ac3127c43195bf53c6c09e9c6fdfff
fff0200ca9a3b0000000017a9146e13fe2f9930597324e9789b62ac27c224e5322d87f0a29a3b0000000017a91475fe0c248a1bb0f3e0280def4d79d55d351f74cb87024730440220 2cfd9ed3f7f95e028b
92b49b4c81f81a48eb632d4ec395afae6d8f65afb2891c02200fdc9514c433ca4df875b09f4d7c6d48578bee1fc5636ba3834691b223134a6d0121027babf4dffa0c3b91b7ff39751bb3bfe35e6c1d80beb
7de977b6152087880063f00000000
Witness Data (response provided by B):
Signature: 304402202cfd9ed3f7f95e028b92b49b4c81f81a48eb632d4ec395afae6d8f65afb2891c02200fdc9514c433ca4df875b09f4d7c6d48578bee1fc5636ba3834691b223134a6d01
Public Key: 027babf4dffa0c3b91b7ff39751bb3bfe35e6c1d80beb7de977b6152087880063f
A -> B Transaction hex: 020000000001015f5a14086b02c21c13301437cb9a437e86fb2edd2f9b3762cacad9b1cf3446b900000000171600145c7e9edd98581f4ac3127c43195bf53c6c09e9c6fdfff
fff020094357700000000017a91475fe0c248a1bb0f3e0280def4d79d55d351f74cb87f06c35770000000017a91411b8cc5383f7a8f977f2aeb640da862c6b80034d87024730440220 0bd69522a47c32770d
5d29d5fdf607117978ebd53d4c5a14d8925b612be6a8c30220657398c5f1121db8a9393915bef2fc8bf75da7c8d59b5b9c6384ee71685ee0d00121020f67d1394038b47013598f75b9c99c048c788e99773
e8a5aa3b3c7675487465a00000000
Challenge Script (ScriptPubKey from A->B tx): OP_HASH160 75fe0c248a1bb0f3e0280def4d79d55d351f74cb OP_EQUAL
----------------------------Transaction Sizes---------------------------------
 A->B Transaction
Transaction Size (in bytes) : 247
Transaction Virtual size: 166
Transaction Weight: 661
 B->C Transaction
Transaction Size (in bytes): 247
Transaction Virtual size: 166
Transaction Weight: 661
```

Segwit 2

```
----------------------------- Final Balances -----------------------------------
DEBUG:BitcoinRPC:-34-> listunspent [1, 9999999, ["2MtrvrYbc6F4NRDsarQV9NyVJUQoU9YyV7C"]]
DEBUG:BitcoinRPC:<-34- [{"txid": "e9386e2fa42775a18fe3ae18c290a67f393c96e4b59e99da31519e5186bfd108", "vout": 1, "address": "2MtrvrYbc6F4NRDsarQV9NyVJUQoU9YyV7C", "
label": "A", "redeemScript": "00145c7e9edd98581f4ac3127c43195bf53c6c09e9c6", "scriptPubKey": "a91411b8cc5383f7a8f977f2aeb640da862c6b80034d87", "amount": 19.9999, "
confirmations": 2, "spendable": true, "solvable": true, "desc": "sh(wpkh([efb8e836/49h/1h/0h/0/1]020f67d1394038b47013598f75b9c99c048c788e99773e8a5aa3b3c7675487465a
))#264w5h71", "parent_descs": ["sh(wpkh([efb8e836/49h/1h/0h]tpubDCLWcmKpcnns5hkiNmjh7zsjYNsv5FffQnvxrAoVw65d46REntNMbkGkUxpLv5XGjgLkG6GvDB1QTfUf3E3gVYBZgCjjLKzMDdG
mFTKrpWd/0/*))#ejpmn66d"], "safe": true}]
Balance of 2MtrvrYbc6F4NRDsarQV9NyVJUQoU9YyV7C: 19.99990000
DEBUG:BitcoinRPC:-35-> listunspent [1, 9999999, ["2N417PFdwWNvRfwEXyCMM5fXYwSCv2yEShU"]]
DEBUG:BitcoinRPC:<-35- [{"txid": "9fce88f76f99e686bc3731fd4c1dbd761c19a8929ff37efcfa7ca00a8bd22d5a", "vout": 1, "address": "2N417PFdwWNvRfwEXyCMM5fXYwSCv2yEShU", "
label": "B", "redeemScript": "001432fdaa8bb21a406be12967b880408cd09498de8d", "scriptPubKey": "a91475fe0c248a1bb0f3e0280def4d79d55d351f74cb87", "amount": 9.9999, "c
onfirmations": 1, "spendable": true, "solvable": true, "desc": "sh(wpkh([efb8e836/49h/1h/0h]tpubDCLWcmKpcnns5hkiNmjh7zsjYNsv5FffQnvxrAoVw65d46REntNMbkGkUxpLv5XGjgLkG6GvDB1QTfUf3E3gVYBZgCjjLKzMDdGm
)#gmkwtgvp", "parent_descs": ["sh(wpkh([efb8e836/49h/1h/0h]tpubDCLWcmKpcnns5hkiNmjh7zsjYNsv5FffQnvxrAoVw65d46REntNMbkGkUxpLv5XGjgLkG6GvDB1QTfUf3E3gVYBZgCjjLKzMDdGm
FTKrpWd/0/*))#ejpmn66d"], "safe": true}]
Balance of 2N417PFdwWNvRfwEXyCMM5fXYwSCv2yEShU: 9.99990000
DEBUG:BitcoinRPC:-36-> listunspent [1, 9999999, ["2N3HGGWxTV3WKVcoxabiZRVKu83ojCAe6AC"]]
DEBUG:BitcoinRPC:<-36- [{"txid": "9fce88f76f99e686bc3731fd4c1dbd761c19a8929ff37efcfa7ca00a8bd22d5a", "vout": 0, "address": "2N3HGGWxTV3WKVcoxabiZRVKu83ojCAe6AC", "
label": "C", "redeemScript": "0014c88745fbf082ca06ae7d3cfd851ee45ff60ce0ea", "scriptPubKey": "a9146e13fe2f9930597324e9789b62ac27c224e5322d87", "amount": 10.0, "con
firmations": 1, "spendable": true, "solvable": true, "desc": "sh(wpkh([efb8e836/49h/1h/0h/0/2]03da6c60589e563890e991a35fb7c35b226e427144d071c0a1acfd14cb2b8f71b7))#
zc9ltslk", "parent_descs": ["sh(wpkh([efb8e836/49h/1h/0h]tpubDCLWcmKpcnns5hkiNmjh7zsjYNsv5FffQnvxrAoVw65d46REntNMbkGkUxpLv5XGjgLkG6GvDB1QTfUf3E3gVYBZgCjjLKzMDdGmFT
KrpWd/0/*))#ejpmn66d"], "safe": true}]
Balance of 2N3HGGWxTV3WKVcoxabiZRVKu83ojCAe6AC: 10.00000000
Balance of A: 19.99990000
Balance of B: 9.99990000
Balance of C: 10.00000000
```

Segwit 3

Verification:

```
sandeep@sandeep-IdeaPad-Slim-3-15IRH8:~/bitcoin/src$ btcdeb --tx=02000000000101018091f663efe9c12ede1ea7e207d3d6ca00f78011f1196bddb1c0bfeadfecc4a0000000017160014e0e706036a871e72044553ca88c0642c23d4199fdfff
ffff0200ca9a3b0000000017a91429b9509ed7393a7a386886e8391c7130723a192b87f0a29a3b0000000017a91485aeef2e3138c54381b0d4345e1d3a94a663afd487024730440220 1d6c0cafc65b0d55de4c5c312b3824f9dbcc09f64566793e065c37ea7ba
9850602202b4dbfd437cced8e3cda250fbf51f4b059190f59f78b43986fa25d65537e2072012102db744fc4dc472bdcec37dbe2305f08feee00dcbbce604e11986fc5a9069980ed00000000 --txin=02000000000101d0c300913224af2a2996487f55d63b4
8df7172a1aae56270fb15312b28aad9da0000000017160014b67690e183dfe7f88bc97dbd6664a2d0d04eaeb0fdffffff020094357700000000017a91485aeef2e3138c54381b0d4345e1d3a94a663afd487f06c357700000000017a9149730f155cf447194be2
c6b7494f71f9d158758c387024730440220251bbae9d8ece9b5dfbc70f19c61fc73fb8c90caf7ec244c9d41f405f65b4482022044e732c0c5ae10b370bd9e1439a956fdfb627ac7e066792da65aea745e16183f012103ccc79c182209bc589364aa99002a931
e00b0a25df4070a220ff42fe659a05bfa00000000
btcdeb 5.0.24 -- type `btcdeb -h` for start up options
LOG: signing segwit taproot
notice: btcdeb has gotten quieter; use --verbose if necessary (this message is temporary)
input tx index = 0; tx input vout = 0; value = 2000000000
got witness stack of size 2
script sig non-empty; embedded P2SH (extracting payload)
hash source = 00141e0e706036a871e72044553ca88c0642c23d4199
22 bytes (P2WPKH)
valid script
- generating prevout hash from 1 ins
[+] COutPoint(4accfeadfe, 0)
note: there is a for-clarity preamble (use --verbose for details)
5 op script loaded. type `help` for usage information
script                                   |                                                                   stack
-----------------------------------------+-------------------------------------------------------------------
OP_DUP                                   | 02db744fc4dc472bdcec37dbe2305f08feee00dcbbce604e11986fc5a9069980ed
OP_HASH160                               | 304402201d6c0cafc65b0d55de4c5c312b3824f9dbcc09f64566793e065c37e...
1e0e706036a871e72044553ca88c0642c23d4199 |
OP_EQUALVERIFY                           |
OP_CHECKSIG                              |
#0000 OP_DUP
btcdeb> step
                [stack] PUSH stack 02db744fc4dc472bdcec37dbe2305f08feee00dcbbce604e11986fc5a9069980ed
script                                   |                                                                   stack
-----------------------------------------+-------------------------------------------------------------------
OP_HASH160                               | 02db744fc4dc472bdcec37dbe2305f08feee00dcbbce604e11986fc5a9069980ed
1e0e706036a871e72044553ca88c0642c23d4199 | 02db744fc4dc472bdcec37dbe2305f08feee00dcbbce604e11986fc5a9069980ed
OP_EQUALVERIFY                           | 304402201d6c0cafc65b0d55de4c5c312b3824f9dbcc09f64566793e065c37e...
OP_CHECKSIG                              |
#0001 OP_HASH160
btcdeb> step
                [stack] POP  stack
                [stack] PUSH stack 1e0e706036a871e72044553ca88c0642c23d4199
script                                   |                                                                   stack
-----------------------------------------+-------------------------------------------------------------------
1e0e706036a871e72044553ca88c0642c23d4199 |                                                                   1e0e706036a871e72044553ca88c0642c23d4199
OP_EQUALVERIFY                           | 02db744fc4dc472bdcec37dbe2305f08feee00dcbbce604e11986fc5a9069980ed
OP_CHECKSIG                              | 304402201d6c0cafc65b0d55de4c5c312b3824f9dbcc09f64566793e065c37e...
#0002 1e0e706036a871e72044553ca88c0642c23d4199
```

Segwit 1

```
btcdeb> step
                [stack] PUSH stack 1e0e706036a871e72044553ca88c0642c23d4199
script                                   |                                                                   stack
-----------------------------------------+-------------------------------------------------------------------
OP_EQUALVERIFY                           |                                                                   1e0e706036a871e72044553ca88c0642c23d4199
OP_CHECKSIG                              |                                                                   1e0e706036a871e72044553ca88c0642c23d4199
                                         | 02db744fc4dc472bdcec37dbe2305f08feee00dcbbce604e11986fc5a9069980ed
                                         | 304402201d6c0cafc65b0d55de4c5c312b3824f9dbcc09f64566793e065c37e...
#0003 OP_EQUALVERIFY
btcdeb> step
                [stack] POP  stack
                [stack] POP  stack
                [stack] PUSH stack 01
                [stack] POP  stack
script                                   |                                                                   stack
-----------------------------------------+-------------------------------------------------------------------
OP_CHECKSIG                              | 02db744fc4dc472bdcec37dbe2305f08feee00dcbbce604e11986fc5a9069980ed
                                         | 304402201d6c0cafc65b0d55de4c5c312b3824f9dbcc09f64566793e065c37e...
#0004 OP_CHECKSIG
btcdeb> step
EvalChecksig() sigversion=1
Eval Checksig Pre-Tapscript
GenericTransactionSignatureChecker::CheckECDSASignature(71 len sig, 33 len pubkey, sigversion=1)
  sig        = 304402201d6c0cafc65b0d55de4c5c312b3824f9dbcc09f64566793e065c37ea7ba9850602202b4dbfd437cced8e3cda250fbf51f4b059190f59f78b43986fa25d65537e207201
  pub key    = 02db744fc4dc472bdcec37dbe2305f08feee00dcbbce604e11986fc5a9069980ed
  script code = 76a9141e0e706036a871e72044553ca88c0642c23d419988ac
  hash type  = 01 (SIGHASH_ALL)
SignatureHash(nIn=0, nHashType=01, amount=2000000000)
- sigversion == SIGVERSION_WITNESS_V0
  sighash    = 85b926af19046b9e6261a567a59b0ea0404ce30685cc16cbe9b187761b505fbe
  pubkey.VerifyECDSASignature(sig=304402201d6c0cafc65b0d55de4c5c312b3824f9dbcc09f64566793e065c37ea7ba9850602202b4dbfd437cced8e3cda250fbf51f4b059190f59f78b43986fa25d65537e2072, sighash=85b926af19046b9e6261
a567a59b0ea0404ce30685cc16cbe9b187761b505fbe):
  result: success
                [stack] POP  stack
                [stack] POP  stack
                [stack] PUSH stack 01
script                                   |                                                                   stack
-----------------------------------------+-------------------------------------------------------------------
                                         |                                                                       01
btcdeb> step
script                                   |                                                                   stack
-----------------------------------------+-------------------------------------------------------------------
                                         |                                                                       01
btcdeb> step
at end of script
```

segwit 2

Part 3: Analysis and Explanation

Comparison of P2PKH (Legacy) and P2SH-P2WPKH (SegWit) Transactions

1. Size comparison of P2PKH and P2SH-P2WPKH transactions

The transaction size was obtained using: `bitcoin-cli -regtest getrawtransaction <txid> true`

The important fields observed were size, vsize and weight.

Metric	P2PKH (Legacy)	P2SH-P2WPKH (SegWit)
Size (bytes)	225	247
Virtual Size (vbytes)	225	166
Weight	900	661

Metric	P2PKH (Legacy)	P2SH-P2WPKH (SegWit)
Fee	Higher	Lower

The SegWit transaction has a smaller virtual size and weight compared to the Legacy transaction.

2. Script structure comparison (challenge-response script)

Feature	P2PKH (Legacy)	P2SH-P2WPKH (SegWit)
ScriptPubKey	`OP_DUP OP_HASH160 <PubKeyHash> OP_EQUALVERIFY OP_CHECKSIG`	`OP_HASH160 <ScriptHash> OP_EQUAL`
scriptSig	`Signature PubKey`	`0 <PubKeyHash>`
RedeemScript	Not used	`0 <PubKeyHash>`
Witness	Not present	`Signature PubKey`
Validation	`scriptSig + scriptPubKey`	`scriptSig + scriptPubKey + redeemScript + witness`
Script result	TRUE	TRUE

- In P2PKH, the signature and public key are stored in scriptSig.
- In P2SH-P2WPKH, the signature and public key are stored in the witness, and scriptSig only contains the redeemScript.

3. Why are SegWit transactions smaller?

- SegWit transactions are smaller because the signature data is separated from the main transaction and stored in the witness field.
- In Legacy transactions, the signature and public key are stored in scriptSig, and the entire scriptSig is counted fully in the transaction size.
- In SegWit transactions, the Signature data is moved from scriptSig to witness, reducing the base transaction size
- Witness data is discounted in weight calculation, so it contributes less to vsize
- scriptSig becomes smaller because it only contains the redeemScript instead of signature + public key
- ScriptPubKey in P2SH-P2WPKH is shorter than the full P2PKH locking script
- The transaction ID does not include witness data, so less data is counted in the base transaction
- Reduced base size lowers the weight, which directly reduces the virtual size
- Smaller virtual size results in a lower fee because fees are calculated per vbyte

weight = base_size × 4 + witness_size vsize = weight / 4

Because the witness part is multiplied by 1 instead of 4, the total weight and virtual size of the SegWit transaction become smaller.

4. Benefits of SegWit transactions

- Lower transaction fees due to smaller virtual size
- More transactions can fit in one block
- Fixes transaction malleability problem
- Improves scalability of the Bitcoin network
- Enables advanced features like Lightning Network